Machine Learning Methodologies for Predicting Fake News on Social Media X: A Comparative Investigation over TruthSeeker Dataset

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***Abstract*— In this era of social media, where every piece of information is at the user's fingertips, anyone who has smart devices and an internet connection can access or create a new piece of information online on these social media sites with just a single click. This changes the delivery of news and how people stay informed. These days, individuals can receive breaking reports on their preferred social media sites without relying on traditional news broadcasting methods. This changes the method by which news is accessed and communicated in the modern internet era. But with this new way of getting information from social media platforms comes the new challenge of determining the credibility and accuracy of the information shared. Most of the time, fake news without proper fact-finding and misinformation can spread quickly, causing confusion and distrust among the public. This kind of fraudulent story is perhaps the foremost risk to any government and may cause social unrest and instability. In this research, our goal is to analyze and predict fraudulent news identification over digital platforms, especially the 'X', also known as 'Twitter, to create a reliable and efficient system that can identify fake news in real-time with high accuracy using machine learning algorithms on the CIC's (Canadian Institute for Cybersecurity's) truth seeker database of 2023. The proposed system is able to recognize fraudulent information across social platforms with its highest 99.97% accuracy using LR and MLP machine learning algorithms, which is much higher as compared to the existing benchmark studies. Further, this proposed system helps to ensure the detection of bogus headlines over various digital media platforms with the highest accuracy, which minimizes the impact of misleading information and safeguards users from being deceived, which helps us to curb the spread of fraudulent breaking news and prevent potential harm to society.**

Keywords*— *Truth seeker, Twitter Dataset, Forged News Discovery, Misinformation, ML, X platform, Digital media.

I. INTRODUCTION

In this era of smartphones and social media, how news reaches us has changed. As per the research published by Paw Research Centre in 2023, among the majority of United States adults, around 86% get the news from tablets and smartphones, which is a 49% increase compared to 2022 and a 51% increase to 2021. In terms of mode of preference to get news, an average US adult prefers digital devices (58%), followed by traditional news mediums like TV (27%), radio (6%), and print (5%). When it comes to which social media site they use for news consumption, Twitter (X) is among the leading 5 online social media platforms with a 12% consumption rate [1]. This rapid shift towards digital news consumption, especially on social media, may raise various concerns, including the reliability and accuracy of the news shared online over these social media platforms. In addition to undermining trust in trustworthy sources of information, fake news or misinformation has numerous detrimental effects on society, such as eroding trust, polarizing and dividing society, compromising public health and safety, political manipulation, having an adverse psychological effect, and having economic repercussions.

According to a 2023 UNESCO and Ipsos worldwide study [2] performed in 16 countries where the elections are lined up for 2024 and discloses the consequences of fake news on political situations in these countries, the study revealed that 56% of users frequently use social media to stay updated on current events. Further, 64% of urban Indians claim that social media is the biggest source of false information, and 87% believe that fake news has already had a serious influence on the political situation in their nation. Further reports published by Social & Media Matters [3] in 2024 in 28 states in India, revealed that 65.2% of first-time voters encountered fake news through mainstream social media sites and 37.1% were directly impacted by this news. The report published by [4] in 2019 reveals that, due to fake news, $78 billion in revenue losses are happening each year.

Social media platforms serve a vital purpose in propagating bogus information due to the convenience of sharing and the lack of credibility for legitimately checking the information shared on these sites. The inherent framework of social media sites promotes effortless distribution of information despite assessing its reliability or fact-checking, unlike traditional media outlets. This creates a favourable environment for the circulation of misinformation and fake news. Furthermore, social media sites concentrate on content based on viewer involvement instead of truthfulness or fact-checking, accelerating the circulation of fake news. As a

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result, viewers are often vulnerable to disinformation, explicitly misleading stories, and propaganda news. On the contrary, the sheer quantity of data posted on these websites makes it challenging for individuals or perhaps government agencies to manually examine the legitimacy and trustworthiness of every news story shared on such websites.

Fake news on social platforms possesses extensive psychological impacts, impacting beliefs, the judgment-making process, democratising, society's beliefs, and cultural standards [5]. Misleading news poses serious threats to democratic discourse, public faith, and potential national safety, drawing the spotlight of experts in recognizing and reducing the effect of bogus reports. To autonomously identify fake news or misinformation, machine-learning techniques play a primary role, as these methods are capable of being trained repeatedly on a massive database that incorporates equally legitimate and counterfeit stories, which determines whether the content released is authentic or not with a superior extent of proficiency. This exploration outlines a framework to assess and anticipate counterfeit headlines over social media platform X (Twitter), employing machine learning techniques and their efficiency assessment for enhanced accuracy.

Further, this article is structured into the subsequent modules, where a literature review is explained in Module II, followed by a research procedure in Module III. The findings and discussion of the study are given in Module IV, and ultimately, recommendations and future objectives are tackled in Module V.

II. Literature Review

M. Iftikhar et al. [6] emphasize real linguistic evaluation and employ machine learning to autonomously recognize fraudulent news reports. However, it could be more efficient if biases in the dataset were resolved and optimized to improve the accuracy of the offered system. In [7], T. Bhatia et al. use the PolitiFact dataset to construct a deep neural network framework by executing features of X (Twitter) members and determining the legitimacy of facts communicated by them to prevent fake news over X (Twitter). However, the provided framework has constraints about flexibility and generalizability. Also, biases in the dataset are not addressed correctly to assure the preciseness and efficiency of the offered system. R. Chauhan [8] uses deep learning, text mining, and NLP (natural language processing) methods to identify counterfeit articles or misinformation on online media sites. Additional FACE Finder and cross-validation are used to improve the accuracy of the system. However, a thorough evaluation of the mentioned methods, encompassing their efficiency and probable drawbacks, is lacking in the offered solution.

A. Narkhede et al. [9], offered a thorough strategy centred on supervised machine learning techniques in conjunction with NLP to recognize misinformation against real news. However, the skewed dataset and shortage of external assurance may affect the functional robustness of the proposed solution. A. Y. Adam et al. [10] present diverse techniques in the offered system to illustrate a versatile and sturdy platform that demonstrates a highly effective and flexible strategy for fake news identification on online platforms. However, there's a shortcoming in adequate experimental confirmation, along with scrutiny of a broader variety of existing fake news identification procedures. In [11], Gupta et al. laid out a three-stage procedure employing numerous machine learning algorithms and highlighting the inclusion of human supervision to boost the prediction's truthfulness. However, the research neglects to investigate the comprehensibility of the offered methodologies.

Sharrab et al. [12] proposed text data forensic instruments and methods employing deep neural networks (DNNs) to identify suspicious behaviours of tweets over the X (Twitter) platform. However, ethical issues in text data investigations might be investigated more thoroughly regarding the ethical consequences of employing DNNs. In [13], S. Agarwal et al. uses SVM-LR and ML techniques for predicting fake news on a novel dataset collected via multiple actual news articles using applications like BS4 and Selenium to optimize the offered model. However, the impact of various news languages, sources, amounts of data, and real-time news feeds is not thoroughly explored in the proposed solution. The study presented by Deepa et al. in [14] employs a passive proactive method for successive knowledge analysis and model updating, combined with natural language analysis strategies to detect misinformation. However, there is a requirement for a more extensive investigation of how effectively the method performs on various databases and in actual world circumstances.

The investigation offered by N. L. S. R. Krishna et al. [15] regarding the efficiency of ML mechanisms, particularly DT and SVM, for discovering fraudulent content related to social media platforms. However, DT and SVM illustrate the potential lack of generalizability of the outcomes. The erratic nature of the data might introduce biases or discrepancies in the prediction models. In [16], M. Aljabri et al. employ several feature mining methods with machine learning techniques on the WELFake dataset to precisely determine fake news. However, the predictability of real-world circumstances is restricted, and it fails to tackle the challenges associated with the volatile characteristics of fraudulent stories. In [17], P. G. et al. examine the vitality of machine learning-centred visual and verbal fake news classification by demonstrating the significance of data sanitizing, encoding, and characteristics extraction.

K. R. Asish et al. [18] employ a machine learning method using Naive Bayes classification to construct a Google Chrome extension centred on US and Indian news to swiftly determine the credibility of news clauses. However, concentrating solely on a particular ML method may not regularly be the most precise or robust approach. Incorporating a wider range of data sets gathered from multiple sources and different languages might improve the tool's flexibility and efficiency in identifying fake news on a more extensive scale. A. Singh et al. [19] explored hybrid models, especially CNN-RNN, SMAN, and transformer-based methods, to enhance effectiveness in identifying false information. However, it could fail to include the hottest innovations or emerging patterns in the discipline of fake news detection. In [20], Z. Tian et al. proposed an approach involving ML methods emphasizing the KNN algorithm and integrating genetic and evolutionary feature selection (GEFeS) and quantitative KNN (QKNN) using ML to identify fake news. However, studies focused on BuzzFace datasets from Facebook in 2016 may not precisely distinguish fake news in the present circumstances and on various online

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platforms. M. Choudhary et al. [21] employ multiple ML methods, such as NB, CNN, LSTM, NN, and SVM, for identifying and minimizing fraudulent news on digital media websites like FB, WhatsApp, & X (Twitter), as well as highlight the continual enhancement of such tools for precise detection.

E. Cueva et al. [22] recommended a framework to recognize fabricated news over Twitter (X) using ML models, especially LSTM, GRU, and NLP. However, problems emerged, specifically as the LSTM and GRU models misclassified satire as real news, while the NLP model successfully detected satire. In [23] A. Zervopoulos et al. offer thorough deep learning techniques for recognizing fake news on X (Twitter) in English and Chinese on a particular dataset utilizing CNNs and LSTM. However, this study is focused exclusively on the Hong Kong protests, which happened in 2019, which might restrict its precision in other contexts, languages, and events. S. Helmstetter et al. [24] present an innovative technique for identifying fake news on X (Twitter) via weakly supervised learning, where source legitimacy acts as a proxy for classifying tweets as fake or non-fake news. However, noisy labels may introduce biases or inaccuracies in the classification process.

The W. Han et al. [25] use ML and DL to deliver comparative assessments of the NB, clustering, and LSTM-CNN hybrid models and demonstrate the significance of choosing the right strategy to achieve a balance between precision and lightness. In [26], N. F. Baarir et al. utilize TF-IDF and n-grams as attribute-gathering methods, with SVM as a classifier for fake news detection due to narrow data sets and processing power.J. Shaikh in [27] employed SVM, NB, and passive-aggressive algorithms to recognize fake news with the greatest precision of 95.05% with the SVM classifier. However, the details regarding dataset utilization in this study are missing.

In the research review, it is noted that the majority of the available research on fake news detection suffers from common limitations like dataset biases, noisy data, improper feature selection, and inadequate comprehensive evaluation.

III. Research Methodology

This section is subsequently structured into subsections, which are carried out in this proposed procedure for fake news detection over X (Twitter) as follows:

A. Dataset

The dataset "TruthSeeker2023" [28] used in this study was generated by the CIC (Canadian Institute for Cybersecurity) in 2023 and is publicly accessible to the research community. Which consists of more than 1,80,000 tagged tweets taken from social media X (Twitter) from 2009 to 2022. While creating this dataset, three-factor active authentication followed by 456 Amazon Mechanical Turkers labelling was used. For enhanced behavioural identification, bots, influence scores, and credibility ratings were included in the dataset. Further, multiple BERT-based methodologies are employed to evaluate the precision of counterfeit news or misinformation detection. The offered dataset in *.csv* file format contains 1,34,198 elements and 64 characteristics for further processing.

B. Data Pre-processing

The data processing procedure serves a vital role in enhancing the precision and attaining a precise outcome of the proposed methodology by carrying out data cleansing, data transformation, attribute selection, data reduction, and dealing with skewness in the dataset. The "TruthSeeker2023" dataset utilized in this study contained precisely 1,34,198 tuples and 64 attributes, among which 68,930 true and 65,268 fake tweets, which are 51.4% and 48.6%, respectively. The pie chart and bar plot representations for the same are represented in Figure 1. It does not hold any missing, null, or duplicate events. Even though the dataset is in an almost balanced state, we employed a minority oversampling strategy to increase the quantity of events in the inferior category comparable to the majority class, i.e., 68930. Further, we use TF-IDF to encode linguistic data into mathematical attributes to utilize in machine learning, which increases the total number of features to 1,064.

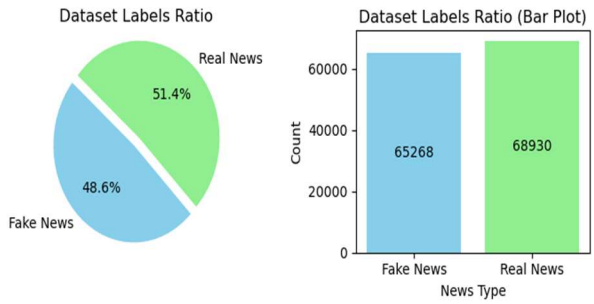

Fig 1. Genuine & Counterfeit news ratio in the truth seeker dataset 2023.

C. Purposed System

The recommended framework in the proposed research is fundamentally divided into information sanitation, and learning modules, as shown schematically in Figure 2.

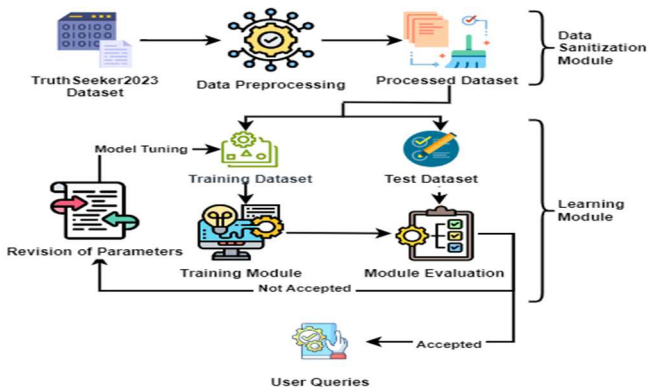

Fig 2. Proposed fake news detection system.

In the data sanitization module, data preparation is performed, in which missing values, null values, noisy data, dimensional reduction, feature selection, and other preprocessing steps, including dataset visualization, are employed to produce a cleaned data set to enhance the efficiency of the recommended framework. In the learning module, the training and test sets are created from the transformed data set in an 80:20 ratio, respectively. Further, several machine learning approaches were developed with the aid of the training set, and subsequently, their precision was checked utilizing the test set. Primarily based on the test set outcomes, the proposed

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approach may be approved or altered via parameter adjustment. For the implementation of the proposed framework, Using an 8th generation Intel i5 CPU, 32 GB of RAM, and 4 GB of NVidia graphic card memory, we prepare Python code by utilizing the capabilities of famous Python libraries, including Pandas, Seaborn, Numpy, Matplotlib, Scipy, and Sklearn.

D. Simulation Parameters

The specifications of the simulation parameters employed in the proposed system throughout this research have been specified in Table I, where *'Dataset Dim.'* represents dataset dimensions, *'S/w or Lib.'* Software or libraries and *'activation 'f''* is activation function.

TABLE I. SIMULATION PARAMETERS USED IN THIS STUDY

Parameter	Value/Setting						
Dataset Name	TruthSeeker2023						
Dataset Dim.	(134198, 64)						
Hardware	i5 2.3GHz CPU,32GB RAM, 4GB NVIDIA GTX1050Ti						
S/w or Lib.	Python, Sklearn, Numpy, Pandas, Matplotlib, Seaborn						
Training Size	107358						
Testing Size	26840						
Algorithm	**LR**	**RF**	**GB**	**MLP**	**DT**	**K-NN**	**NB**
k-Fold	10	10	10	10	10	10	10
Learning Rate	N/A	N/A	0.1	.001	N/A	N/A	N/A
Epochs	1000	No	Not set	2000	Not set	Not set	Not set
Optimizer	liblinear	N/A	GD	Adam	N/A	N/A	N/A
Regularization	1	100	100	.0001	1	5	1E-09
Activation 'f'	Sigmoid	None	None	ReLU	None	None	None
Random Seed	N/A	42	42	42	42	N/A	N/A

IV. RESULT & DISCUSSION

In this module, the outcomes of the Truth Seeker dataset 2023 have been addressed through data visualization and machine learning classification techniques demonstrated as follows:

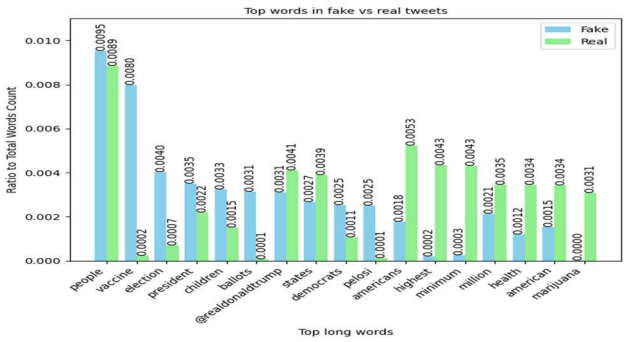

Fig 3. Most occurred long words used in fake & real news.

Figure 3 indicates that *'people'* is the most regularly used lengthy word, with 0.0095 and 0.0089 ratios to the false and real word counts, respectively. Followed by *'vaccination'*, with around 0.0080 fake and 0.0002 in actual news tweets. Followed by the *'election'* with 0.0040 and 0.0007, respectively, for both false and actual tweets, and similar.

Table 2 presents a vital understanding of the different linguistic and contextual characteristics associated with fake news. The aforementioned metrics, including average word length, BotScore, and event percentage, offer beneficial statistical metrics for determining counterfeit news content. It's also essential to consider that fraudulent news demonstrates a greater regularity of specific linguistic attributes corresponding to real news, including adverbs and conjunctions. This discovery points out the habit of counterfeit news to employ a higher level of sentimentality and emotional narrative style. Moreover, characteristics like BotScore and normalize_influence show the probable appearance of automated or bot-generated content, leading to the propagation of fake news. By introducing these distinctive cognitive patterns into evaluation frameworks, the precision and reliability of the detection systems may increase.

TABLE II. ESSENT. ATTRIBUTE EVALUATIONS FOR IDENTIFYING FAKE NEWS

Feature	Fake News Average	Fake News Exact
Average word length	5.017788791	327501.0388
BotScore	0.061158025	3991.662
EVENT_percentage	0.001877022	122.5094951
FAC_percentage	0.004215029	275.1065369
MONEY_percentage	0.021043313	1373.454948
Max word length	13.10269964	855187
ORG_percentage	0.210929861	13766.97018
PERCENT_percentage	0.026046792	1700.021995
PERSON_percentage	0.256112315	16715.93855
Word count	36.1944138	2362337
adjectives	2.925108782	190916
adverbs	1.654823191	108007
ampersand	0.123812588	8081
conjunctions	1.054605626	68832
followers_count	5128.313308	334714753
listed_count	35.67434271	2328393
normalize_influence	0.074116251	4837.419502
questions	0.342357664	22345
replies	2.57619354	168143
short_word_freq	22.30040755	1455503
unique_count	2.265873016	147889

Conversely, based on Table 3 evaluation, it provides a thorough review of essential characteristics critical for recognizing real news. With an estimated BotScoreBinary of 0.0301 and an accurate count of 2077, it demonstrates a comparatively lesser tendency regarding automated operations in comparison to other attributes. Furthermore, characteristics like CARDINAL_percentage, DATE_percentage, and LANGUAGE_percentage reveal noteworthy ratios of numeric, temporal, and semantic components in genuine news articles, each of which averages approximately 0.1. Additionally, the majority of TOs and URLs, with a mean of 0.7400 and 0.6793, respectively, indicate the noteworthy existence of immediate addresses and URLs. Further evaluation indicates a significant emphasis on adpositions, pronouns, and favourites, demonstrating the relevance of linguistic frameworks and consumer desires for genuine news content. The utilization of said real news component features can significantly impact the efficiency of detection systems in finding and filtering out fake news.

TABLE III. ESSENT. ATTRIBUTE EVALUATIONS FOR IDENTIFYING REAL NEWS

Feature	Real News Average	Real News Exact
BotScoreBinary	0.030132018	2077
CARDINAL_percentage	0.105148666	7247.89754
DATE_percentage	0.107374426	7401.319189
LANGUAGE_percentage	0.001742352	120.1003283
TIME_percentage	0.004715118	325.0131036
TOs	0.740040621	51011
URLs	0.679326853	46826
adpositions	3.668199623	252849
favourites	20.86656028	1438332
favourites_count	35931.9863	2476791816
pronouns	1.313695053	90553
quotes	0.366705353	25277
statuses_count	39876.35287	2748677003

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Figure 4 demonstrates a word, text, or tag cloud made using fake news tweets that illustrates the significant concerns and patterns addressed in the fake news tweets, since the scale of every phrase is corresponding to its repetition or impact of the text. It brings out the foremost crucial and prominent keywords that appear in the false reports.

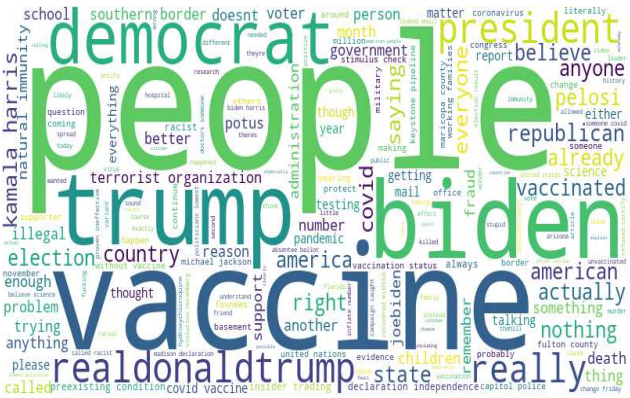

Fig 4. Fake news tweets word cloud

The efficacy evaluation of numerous machine learning classification models, including LR, RF, GBM, MLP, DT, KNN, and NB, was carried out to determine the most appropriate approach to fake news detection. Table 4 illustrates the same, where 'MN' represents the model name, accuracy is represented by 'ACC'. Precision, recall, F1 score, and ROC-AUC curve are represented by 'PRE' , 'REC', 'F1-S', and 'R-A', respectively. In addition to the 'ET (s)' representing the execution time in (seconds) and 'MR (MB)' representing the memory required in (MB) of every classification procedure concerning the dataset used on the aforementioned hardware.

TABLE IV. PERFORMANCE EVALUATION OF ML CLASSIFICATION MODELS

MN	ACC	PRE	REC	F1-S	R-A	ET (s)	MR (MB)
LR	0.9997	0.9997	0.9998	0.9997	1	9.27	8571.57
RF	0.9984	0.9992	0.9976	0.9984	0.9999	108.9	8590.87
GB	0.9735	0.9793	0.9677	0.9734	0.9969	199.48	8597.2
MLP	0.9997	0.9997	0.9998	0.9997	1	56.27	8602.31
DT	0.9989	0.9994	0.9983	0.9989	0.9989	70.96	8603.24
K-NN	0.999	0.9998	0.9983	0.999	0.9999	52.02	8609.2
NB	0.9624	0.9767	0.9477	0.962	0.9837	2.2	8609.25

The systematic study of the proposed methods demonstrates that each technique has upsides and downsides for various assessment standards. Logistic Regression (LR) and Multilayer Perceptron Classifier (MLP) come out as the strongest competitors, both demonstrating outstanding accuracy of 0.9997. LR exhibits marginally superior recall (0.9998), F1 Score (0.9997), execution time (9.27 seconds), and memory (8571.57 MB) as compared to MLP (0.9983), (0.9990), (52.02 seconds), and (8609.20 MB), respectively, while MLP exhibits a slightly greater ROC AUC (0.9998). Decision Tree (DT) and K-Nearest Neighbours (K-NN) deliver commendable accuracy (0.9989 and 0.9990, respectively) with significantly limited memory utilization (8590.87 and 8597.20 MB, respectively) but with significantly greater computation times (108.90 and 199.47 seconds, respectively).

Random Forest (RF) also showcases impressive accuracy (0.9984), precision, recall, and F1 score (0.9997), albeit with considerably moderate memory usage (8602.31 MB) and execution times (56.27 seconds). The Gradient Boosting Machine (GBM) attains adequate accuracy (0.9734) and precision (0.9994) but falls inferior in terms of higher memory usage (8603.24 MB) and execution time (70.96 seconds). Naive Bayes (NB) demonstrates greater precision (0.9767) and recall (0.9477) but performs poorly in terms of accuracy (0.9624) and highest memory utilization (8609.24 MB) across all methods, but with the lowest execution time (2.20 seconds). Overall, examining a harmony among efficiency metrics, processing time, and memory usage, Logistic Regression (LR) and Multilayer Perceptron Classifier (MLP) come out as the most efficient choices for this identification objective. The ROC curve for the aforementioned techniques is depicted in Figure 5.

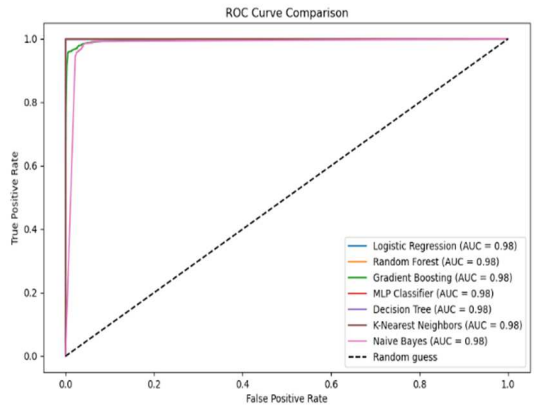

Fig 5. ROC curve for ML Classification Models

Moreover, Table 5 demonstrates the proposed system prediction accuracy evaluations with the existing benchmark studies, particularly those that leveraged ML methodologies to determine fraudulent reports and are pertinent to our research. Where 'BSFNPA' represents benchmark study fake news prediction accuracy and 'PSFNPA' represents proposed study fake news prediction accuracy.

V. CONCLUSION & FUTURE SCOPE

In the proposed study, based on examining and comparing the efficacy assessments of different machine learning approaches, LR (logistic regression) emerges as the top-performing approach, demonstrating the greatest precision, accuracy, and F1 score of 0.9997. Following LR, the Multilayer Perceptron Classifier (MLP) illustrates outstanding performance concerning an accuracy and precision of (0.9997), an F1 score of (0.9990), and K-Nearest Neighbours (K-NN) tightly trailing MLP with an accuracy of 0.9990. Decision Tree (DT) exhibits competitive efficacy with an accuracy of 0.9988. That demonstrates the proposed framework actually performed exceptionally well, which exhibits the superior capability to distinguish the fake news over the social media platform 'X'.

TABLE V. COMPARISON OF PROPOSED FAKE NEWS PREDICTION SYSTEM AGAINST BENCHMARK STUDIES

Study Ref.	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]	[14]	[15]	[16]	[18]	[20]	[22]	[23]	[24]	[25]	[27]	[28]
BSFNPA	92%	91.11%	94%	90%	85%	94%	96%	88%	96%	91.74%	98%	93.22%	87%	99%	99.30%	90%	82.29%	95.05%	70.01%
PSFNPA	99.97%																		

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Further, the envisioned framework would possibly be enhanced in the foreseeable future by integrating advanced methodologies such as hybrid ML and DL procedures to elevate the accuracy and precision. Furthermore, our subsequent endeavours will include augmenting our dataset with different data obtained from diverse social media platforms by improving the algorithm's capability to identify and categorize false facts, covering a broader range of online content to more efficiently tackle the expansion of counterfeit reports and disinformation on different social media platforms.

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